1 Introduction

The purpose of the Lab 2 is to provide understanding about how to read data from analog inputs and presenting three distinct experiments aimed at deepening our practical understanding about Arduinos. This session starts with construction of a voltmeter and an ampermeter using Arduino, provides information about how these instruments can be integrated into circuits. Moreover, the introduction of the CNY70 Sensor with Arduino on a breadboard deepens our understanding into real-world sensor applications.

2 Experiment

In this laboratory session we will held three different experiments.
First two experiments will provide better understand how we can construct voltmeter and ampermeter with Arduino on a simple circuit and last experiment will cover how to use CNY70 Sensor with Arduino on breadboard.

2.1 Experiment 1: Voltmeter with Arduino

In our first experiment we will construct a voltmeter with using Arduino.
First we will construct the voltmeter on the breadboard then we will upload the essential codes for to measure voltage into the Arduino. With our constructed voltmeter we will able to measure voltage with analog inputs of Arduino and we will display the voltage on the console of Arduino.

For the first experiment we will construct the circuit shown below in the Figure 1.

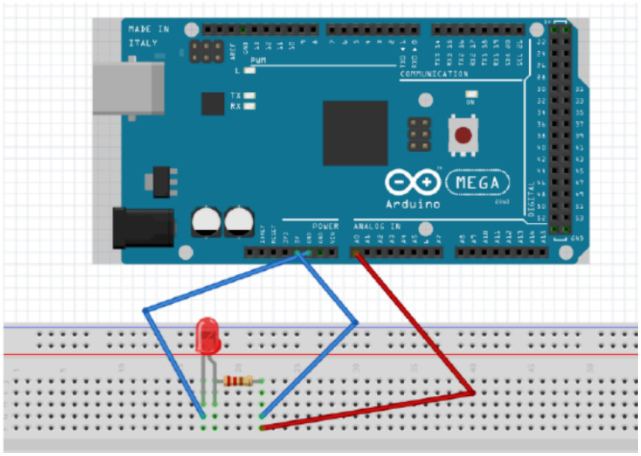


Figure 1: Arduino voltmeter circuit scheme

In this experiment we used these components listed below.

- 1X Arduino Mega with USB cable to connect Arduino to computer
- 1X 220kΩ resistor
- 1X LED
- 1X Breadboard
- 3X Jumper cables

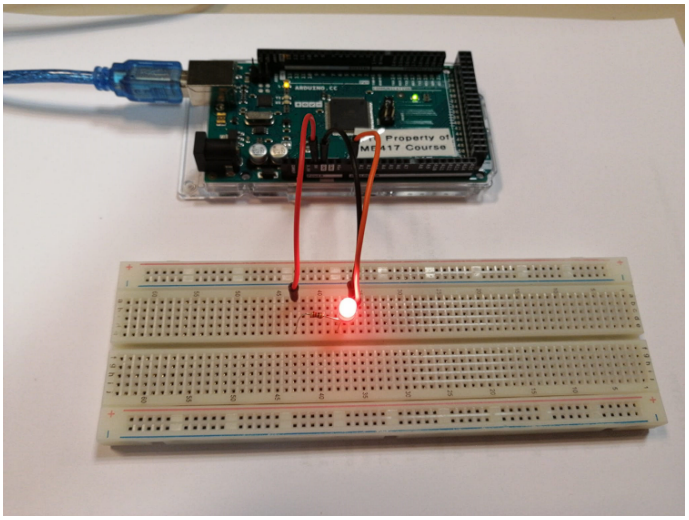


Figure 2: First step of Voltmeter with Arduino

As we can see from the Figure 2 shown above, +5V output from the Arduino is connected to positive side of the LED (where is the resistor connected to the LED) and GND is connected to the negative leg of the LED.

Normally, on the handout of the laboratory it is stated to measure voltage values between each leg of the resistor but voltage drop over the resistor can be calculated with Kirchhoff's Voltage Law, so we have measured voltage values of each leg of LED.

In order to get measurements we connected A0 analog input to negative leg of the LED the as shown in the Figure 2. With the analog input A0, we can read the data on Arduino.

To read the data coming from A0 analog input, we had to upload some codes to Arduino board because Since Arduino Mega's analog inputs have 10-bit resolution it maps 0-5V input voltage into integer values between 0 and 1023. So, the read input must be multiplied by $5/1023 = \approx 0.00488$ in order to have the real voltage value. As shown in the Figure 3 one we have implemented some codes to Arduino, verified the code and uploaded to Arduino as we have experienced in the laboratory 01. In the console we see the voltage values are measured as 0.00V in the red rectangle area.



Figure 3: Voltmeter code in Arduino and voltage values of first step

We will implement the same approach for the positive leg of the LED, we have connected the A0 analog input to the positive leg of the LED as shown in the Figure 4 and run the same code on Arduino to measure voltage value of the positive leg of LED.

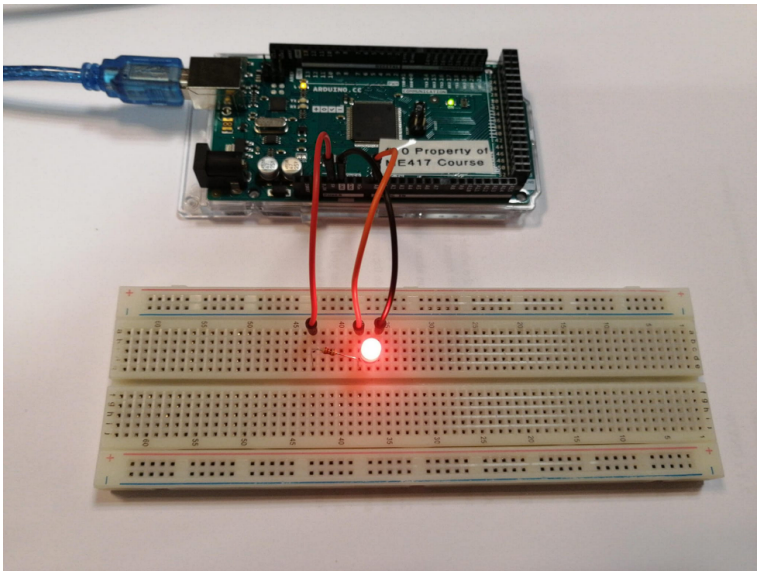


Figure 4: Second step of Voltmeter with Arduino

In the Figure 5 below here, we can see the measured value for the positive leg of the LED is measured as 2.01V.

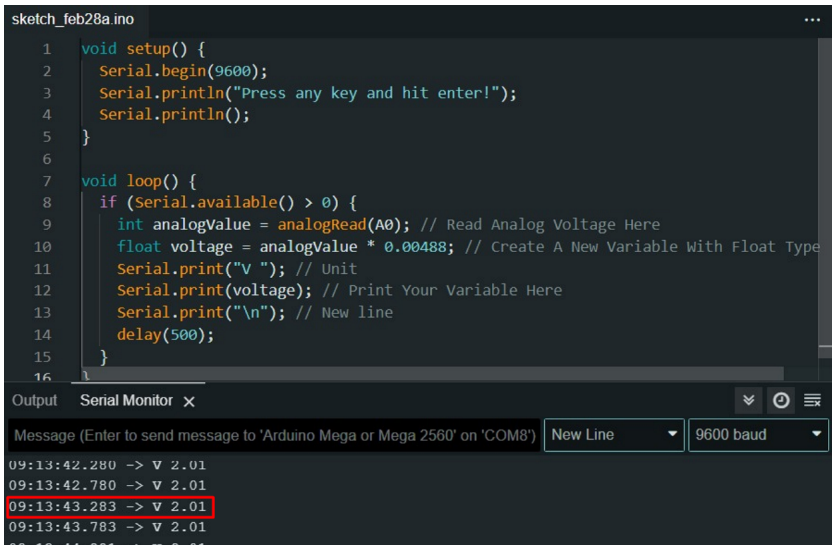


Figure 5: Voltmeter code in Arduino and voltage values of second step

We know the supplied voltage (V_S) is +5V and the voltage drop over the LED (V_{LED}) is measured alread as 2.01V, the voltage drop over the 220Ω resistor (V_R) can be calculated using Kirchhoff's Voltage Law.

$$V_S = V_{LED} + V_R -> 5V = 2.01V + V_R -> V_R = 2.99V$$

2.2 Experiment 2: Ampermeter with Arduino

In our second experiment we will construct a ampermeter with using Arduino. First we will construct the ampermeter on the breadboard then we will upload the essential codes as we have done in the first experiment. With our constructed ampermeter, we will able to measure current flow through the points we have decided with using analog inputs A0 and A1 of Arduino and we will display the how much current passing through the specific nodes we connect A0 analog input to the breadboard on the console of Arduino.

For the first experiment we will construct the circuit shown below in the Figure 6.

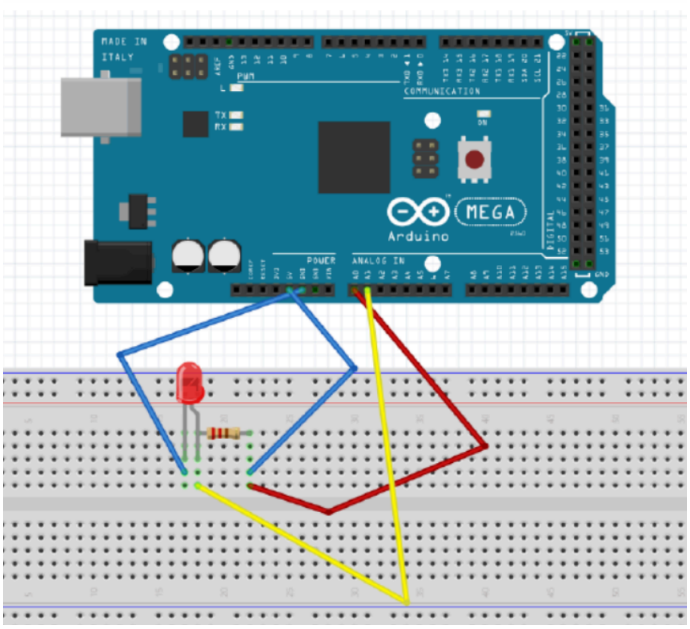


Figure 6: Arduino ampermeter circuit scheme

In this experiment we used these components listed below.

- 1X Arduino Mega with USB cable to connect Arduino to computer
- 1X 220kΩ resistor
- 1X LED
- 1X Breadboard
- 4X Jumper cables

As shown in the Figure 7, we have built the circuit on the breadboard and connected +5V with red cable to resistors leg, GND with black cable to LED's leg, A0 analog input to resistors leg (same position with +5V input), and A1 analog input to negative leg of the LED (same position with GND connection).

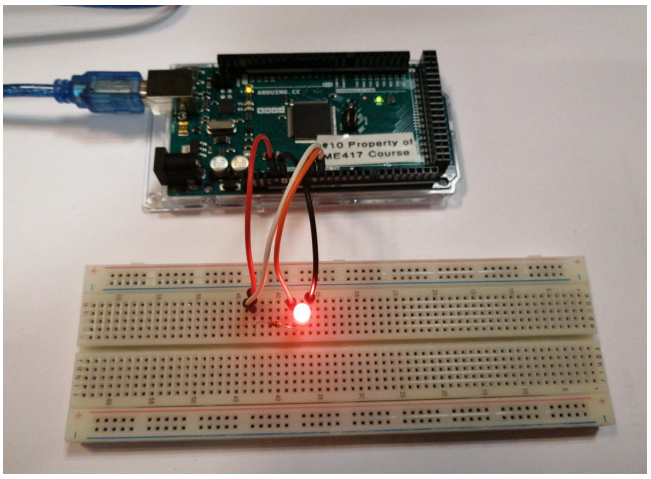


Figure 7: Ampermeter with Arduino

The difference between the two analog pin readings is taken by the code and divided by 220. After that, Amps(A) are converted to milliAmps(mA) by multiplying the result by 1000. Figure 8 displays the code and the measurement result. The Arduino's analog inputs map 0–5V input voltage into integer values between 0 and 1023 because of its 10-bit resolution. For the real voltage and current values, multiply the read input by $5/1023 = \approx 0.00488$.

As we can see from the Figure 8, in the red rectangle our measurements shows -13.53mA ("-" sign is due to sign convention). Then we check our measurements with applying Ohm's Law. We know the resistor value and in the part 2.1 Experiment 1, we have found that voltage drop over the resistor was 2.99V. If we apply Ohm's Law with these values, we get this equation below.

$$2.99V/220\Omega = I = 0.0136A = 13.59mA \tag{1}$$

Our calculation (13.59mA) is close enough to our measurements (13.53mA) so we can claim that, our ampermeter works as it should be.

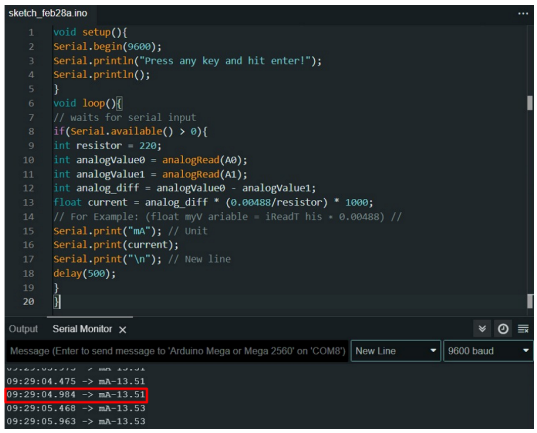


Figure 8: Ampermeter code and measurements

2.3 Experiment 3: CNY70 Sensor

CNY70 sensor has an infrared receiver and transmitter. This sensor is used to distinguish black and white colors from 2-3 cm away. The infrared ray sent by the transmitter is reflected from a surface and returns to the receiver.

In our experiment we will use CNY70 sensor and Arduino to identify if the object is dark or light coloured.

Firstly we will construct the circuit shown below in the Figure 9.

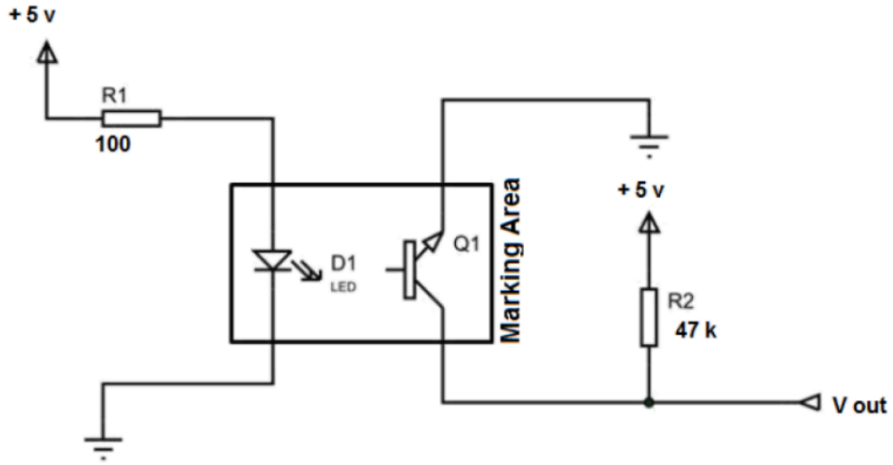


Figure 9: CNY70 sensor circuit scheme

To construct circuit shown in the Figure 9, we used these components listed below.

- 1X Arduino
- 1X CNY70 sensor
- 1X Dark colored object (we used pen)
- 1X 100Ω resistor
- 1X 220Ω resistor
- 1X 47kΩ resistor
- 1X LED
- 1X Breadboard

In the Figure 10 you can see the constructed circuit with CNY70. Red cable is +5V, black cable is GND and white cable reads output from the CNY70 sensor.

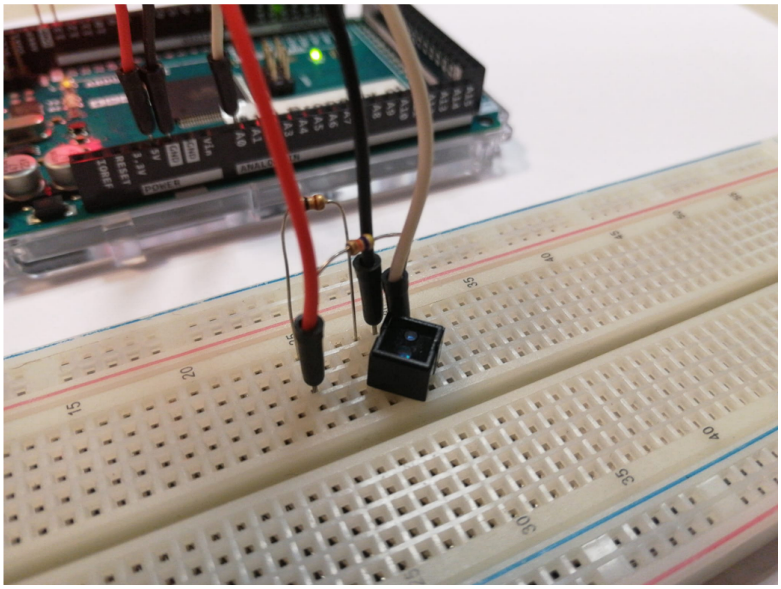


Figure 10: CNY70 sensor circuit

In order to operate the circuit we need an if-else loop implemented to the circuit. When the sensor voltage reading exceeds a pre-determined voltage level, the loop writes HIGH to digital pin 13 which is LED is connected to or setting the digital pin 13 output to LOW otherwise. By monitoring the voltage reading on the serial monitor while blocking and unblocking the sensor, that "certain value" was discovered.

sketch_feb28a.ino

```
1  void setup() {
2    Serial.begin(9600);
3    Serial.println("Press any key and hit enter!");
4    Serial.println();
5    pinMode(13, OUTPUT);
6  }
7
8  void loop() {
9    if (Serial.available() > 0) {
10     int analogValue = analogRead(A0); // Read Analog Voltage Here
11     float voltage = analogValue * 0.00488; // Create A New Variable With Float Type
12     Serial.print("V "); // Unit
13     Serial.print(voltage); // Print Your Variable Here
14     Serial.print("\n"); // New line
15     delay(200);
16     if (voltage > 0.2) {
17       digitalWrite(13, HIGH);
18     }
19     else{
20       digitalWrite(13, LOW);
21     }
22   }
23 }
```

Figure 11: CNY70 sensor circuit code implemented

As shown in the Figure 12, we can observe the circuit works and LED lights up while sensor is not detecting anything.

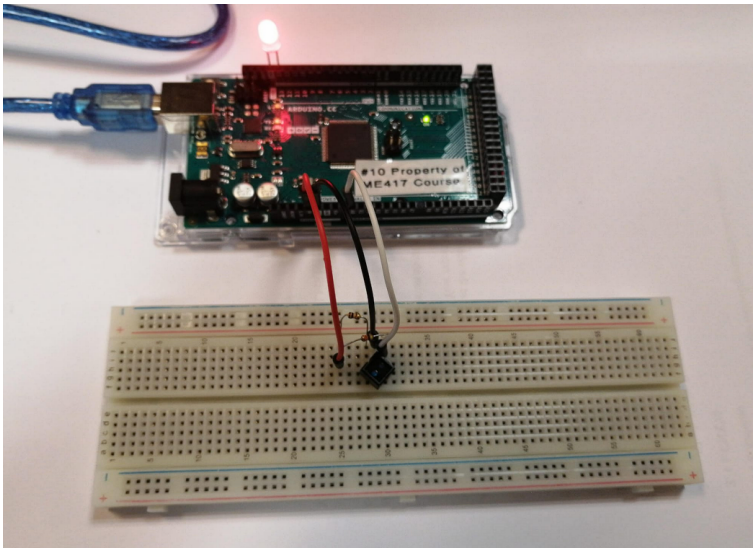


Figure 12: CNY70 sensor is not detecting anything and lights up

But when the sensor detects anything in dark colored (in our case it is dark brown colored pen) LED lights off as shown in the Figure 13.

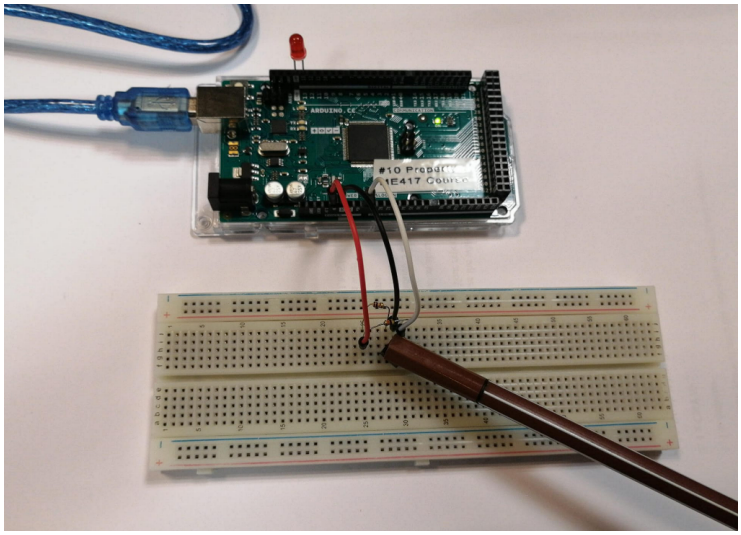


Figure 13: CNY70 sensor is detecting the pen and lights down

3 Conclusion

Lab 2 significantly advanced our practical understanding of Arduino applications, focuses on analog input readings. Through constructing a voltmeter, an ampermeter, and using the CNY70 Sensor for object detection, we explored some of the Arduinos' capabilities.

In the experiment we have used the Kirchhoff's Voltage Law, coded the Arduino for accurate readings and read digital inputs. These hands-on experiences are helpful for our mechatronics knowledge, enhancing our understanding in applying Arduino in various electronic and sensing scenarios.